Chapter 13: Sampling Distributions

The Practice of Statistics in the Life Sciences
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Part III: Foundations of Statistical Inference

Why This Chapter Matters

Statistical inference uses information from a **sample** to learn about a larger **population**.

However, different random samples from the same population generally produce different values of a statistic. This natural sample to sample variation is called **sampling variability**.

Central Idea

Chapter 13 examines how statistics vary across repeated random samples. Understanding this variation allows us to judge how reliably a sample statistic estimates a population parameter.

By the end of the chapter, you should be able to answer the following questions:

- What is the difference between a parameter and a statistic?
- What is a sampling distribution?
- Why do larger samples generally produce more stable statistics?
- When can a Normal model be used for a sample mean or a sample proportion?

Parameters and Statistics

A **parameter** is a numerical characteristic of a population.

- It is fixed for a particular population
- It is often unknown because measuring the entire population may be impractical

A **statistic** is a numerical characteristic calculated from sample data.

- It can be calculated once the sample has been observed
- Its value may change when a different random sample is selected

Measure	Sample statistic	Population parameter
Mean	$\bar{x}$	μ
Proportion	$\hat{p}$	p

A statistic is used to estimate or learn about the corresponding population parameter.

Three Different Distributions

Three related but distinct distributions appear throughout this chapter.

Population Distribution

The distribution of a variable for all individuals in the population

Sample Distribution

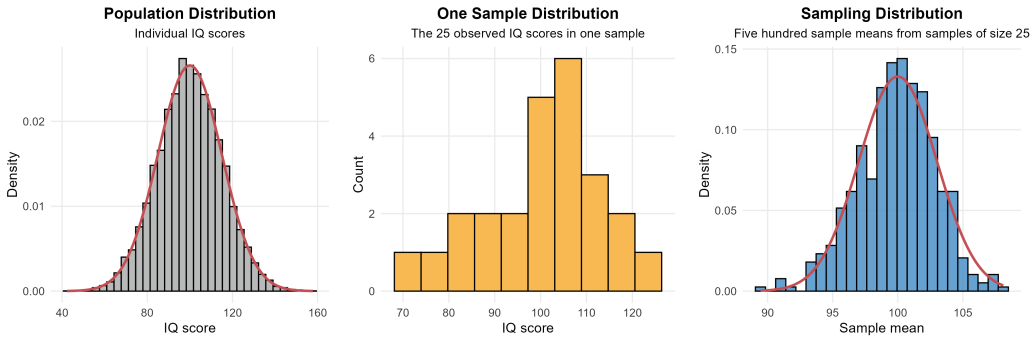
The distribution of the observed values for individuals in one particular sample

Sampling Distribution

The distribution of a statistic calculated from all possible random samples of the same size from the population

Note: Population and sample distributions describe **individual observations**. A sampling distribution describes the possible values of a **statistic**.

Connecting the Three Distributions



We begin with a population, select a random sample and calculate a statistic. Repeating this process many times produces the sampling distribution of that statistic.

Example 13.1: Identifying Population Parameters

Question

A national registration system contains complete records for every birth in the population of interest during one year. Among all registered births, 8% were classified as low birth weight. Among all people giving birth for the first time during that year, the mean age was 26.3 years. Are these values parameters or statistics? Give the appropriate notation.

Solution

Both values describe entire populations defined by the study rather than samples. Therefore, both are **parameters**.

The population proportion of low-birth-weight births is $p = 0.08$

The population mean age at first birth is $\mu = 26.3$

Whether a value is a parameter or a statistic depends on where the data came from, not on the size of the numerical value.

Example 13.2: Identifying Sample Statistics

Question

A polling organization selects a random sample of 1023 adults. In the sample, 19% report smoking cigarettes during the previous week. Among the sampled adults who reported smoking, the mean number of cigarettes smoked per day is 12. Are these values parameters or statistics? Give the appropriate notation.

Solution

Both values were calculated from sample data. Therefore, both are **statistics**.
The sample proportion reporting cigarette use is $\hat{p} = 0.19$
The sample mean number of cigarettes smoked per day among the sampled smokers is $\bar{x} = 12$

These statistics describe the observed sample and may be used to estimate corresponding parameters in the target population.

Repeated Sampling and the Sampling Distribution

A statistic is not fixed before a random sample is selected. Its value depends on which individuals happen to be included in the sample.

If we repeatedly select random samples of the same size, the value of the statistic will vary from sample to sample.

Sampling Distribution

The sampling distribution of a statistic is the probability distribution of that statistic over all possible random samples of the same size selected using the same sampling process.

Note: The observations in a sampling distribution are **statistics calculated from samples**, not measurements from individual members of the population.

Examples 13.3 and 13.4: Simulating Sample Means

Setting

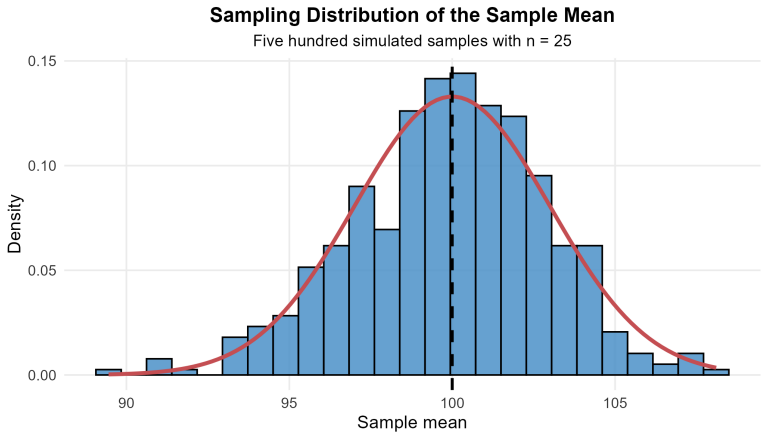
Individual IQ scores are modelled by a Normal distribution with $\mu = 100$ and $\sigma = 15$. A resaercher uses R to select 500 random samples, each containing $n = 25$ IQ scores.

For each simulated sample, the following process is used:

1. Select 25 IQ scores from the population
2. Calculate the sample mean $\bar{x}$
3. Record the value of $\bar{x}$
4. Repeat the process until 500 sample means have been obtained

The 500 recorded sample means approximate the sampling distribution of $\bar{x}$ for samples of size 25.

Examples 13.3 and 13.4: Simulating Sample Means



Examples 13.3 and 13.4: Simulating Sample Means

The simulated sampling distribution has three important features:

- It is centred near the population mean of 100
- It has an approximately Normal shape
- It has considerably less spread than the distribution of individual IQ scores

Individual IQ scores have a standard deviation of 15, but the 500 sample means have a standard deviation close to 4. Sample means vary less than individual observations.

The Sampling Distribution of $\bar{x}$

Suppose a random sample of size n is selected from a population with mean μ and standard deviation σ.

Centre

The mean of the sampling distribution equals the population mean

$$\mu_{\bar{x}} = \mu$$

Therefore, $\bar{x}$ is an **unbiased estimator** of μ. Unbiased does not mean that every sample mean equals μ. It means that the sample means are centred at μ over repeated random samples.

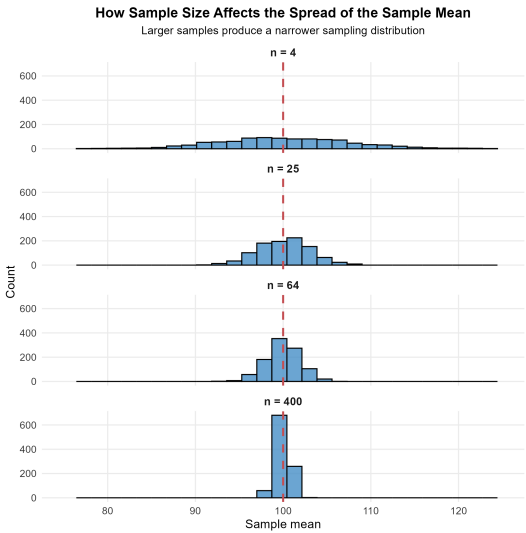
Spread

The standard deviation of the sampling distribution is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

This quantity describes the typical sample-to-sample variation in $\bar{x}$.

How Sample Size Affects the Sample Mean



As the sample size increases:

- The centre of the sampling distribution remains at μ
- The spread of the sampling distribution decreases
- The sample mean is more likely to fall close to the population mean

Main Idea

Larger random samples produce more stable and more precise estimates of the population mean.

Why the Square-Root Relationship Matters

The standard deviation of the sample mean is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Therefore, the improvement in precision depends on the **square root** of the sample size. If the sample size is multiplied by 4, then

$$\frac{\sigma}{\sqrt{4n}} = \frac{1}{2}\frac{\sigma}{\sqrt{n}}$$

The standard deviation of $\bar{x}$ is divided by 2. If the sample size is multiplied by 100, then

$$\frac{\sigma}{\sqrt{100n}} = \frac{1}{10}\frac{\sigma}{\sqrt{n}}$$

The standard deviation of $\bar{x}$ is divided by 10.

Note: To reduce the standard deviation of $\bar{x}$ by a factor of c, the sample size must be multiplied by c^2. Larger samples improve precision but the improvement becomes increasingly expensive.

When Is the Sampling Distribution Normal?

The shape of the sampling distribution of $\bar{x}$ depends on both the shape of the population distribution and the sample size.

1. The Population is Normal: If the population distribution is Normal, then the sampling distribution of $\bar{x}$ is **exactly Normal** for every sample size n.
2. The Population is not Normal: If the population distribution is not Normal, the Central Limit Theorem tells us that the sampling distribution of $\bar{x}$ becomes **approximately Normal** as the sample size increases.

There is no single sample-size requirement that works for every population. Strong skewness and extreme outliers generally require a larger sample.

Example 13.5: Population Values and Sample Means

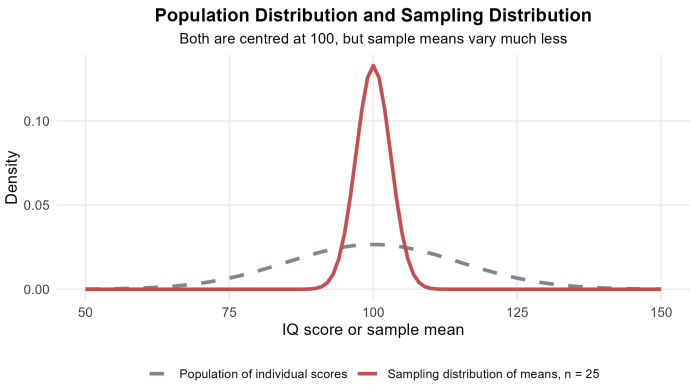
Question

Individual IQ scores are modelled by a Normal distribution with mean $\mu = 100$ and standard deviation $\sigma = 15$. Random samples of size $n = 25$ are selected. How does the population distribution of individual IQ scores compare with the sampling distribution of $\bar{x}$?

Feature	Individual IQ score	Sample mean $\bar{x}$
Centre	$\mu = 100$	$\mu_{\bar{x}} = 100$
Standard deviation	$\sigma = 15$	$\sigma_{\bar{x}} = \frac{15}{\sqrt{25}} = 3$
Shape	Normal	Exactly Normal

The two distributions have the same centre and the same general shape, but the sample means have much less variability than individual IQ scores.

Example 13.5: Population Values and Sample Means



Individual IQ scores

- Centred at 100
- Standard deviation of 15
- Considerable variation among individuals

Sample means for $n = 25$

- Also centred at 100
- Standard deviation of 3
- Much more concentrated around the population mean

Note: Averages are less variable than individual observations.

The Central Limit Theorem

Central Limit Theorem

Suppose independent observations are selected from a population with finite mean μ and finite standard deviation σ. As the sample size n increases, the sampling distribution of $\bar{x}$ becomes approximately Normal, with

$$\mu_{\bar{x}} = \mu \qquad \text{and} \qquad \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

What the Theorem Says

The distribution of **sample means across repeated samples** becomes more nearly Normal as n increases.

What It Does Not Say

The theorem does not say that the population or the observations within one large sample become Normal.

Example 13.6: The Central Limit Theorem in Action

Question

Suppose a population distribution is strongly right-skewed. How does the sampling distribution of $\bar{x}$ change as the sample size increases?

A sample mean combines information from several observations. As more observations are included in each mean, the effect of individual unusually large values becomes less dominant. As the sample size increases

- The sampling distribution becomes less right-skewed
- Its shape becomes more nearly Normal
- Its centre remains at μ
- Its standard deviation decreases according to

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

The population remains right-skewed. It is the sampling distribution of $\bar{x}$ that becomes approximately Normal.

Assessing Whether a Normal Approximation Is Reasonable

There is no universal minimum sample size that guarantees an appropriate Normal approximation for every population. The assessment should consider the following features together:

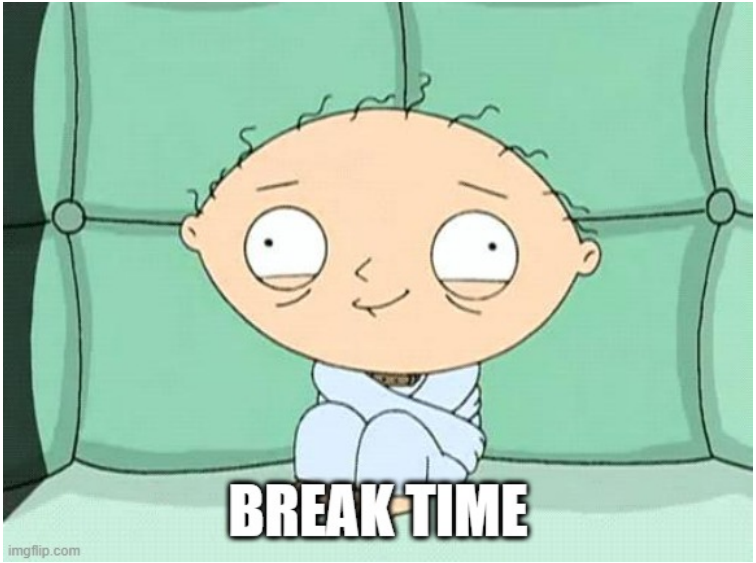
- The available information about the shape of the population distribution
- The size of the random sample
- The presence of substantial skewness or unusual observations

General Guidance

- A population that is approximately symmetric and has no substantial outliers may require less averaging
- A population with substantial skewness or outliers may require a larger sample before the sampling distribution of $\bar{x}$ is approximately Normal

Sample size should not be considered by itself. The shape of the population distribution also affects how quickly the Normal approximation becomes reasonable.

Break Time



The Sample Proportion

For a categorical variable with two possible outcomes, the statistic of interest is often the **sample proportion**. Let

- n represent the sample size
- X represent the number of observations in the category of interest

The sample proportion is

$$\hat{p} = \frac{X}{n}$$

Population and Sample Notation

- p represents the proportion in the population
- $\hat{p}$ represents the proportion observed in the sample

The Sampling Distribution of $\hat{p}$

Suppose a random sample of size n is selected from a population in which the proportion belonging to the category of interest is p.

Centre: The mean of the sampling distribution is

$$\mu_{\hat{p}} = p$$

Spread: The standard deviation is

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

Conditions: The observations should be obtained through an appropriate random process. When sampling without replacement, a common condition is

$$n \leq 0.10N$$

Normal Approximation: The sampling distribution of $\hat{p}$ is approximately Normal when

$$np \geq 10 \qquad n(1-p) \geq 10$$

Example 13.8: Contact Lens Practices

Question

A random sample of 281 contact-lens users contains 139 people who report that they do not consistently wash their hands before handling their lenses. What is the sample proportion?

Define a success as a sampled person reporting that they do not consistently wash their hands before handling their contact lenses. The number of successes and the sample size are

$$X = 139 \qquad \text{and} \qquad n = 281$$

Therefore

$$\hat{p} = \frac{X}{n} = \frac{139}{281} \approx 0.495$$

Approximately 49.5% of the people in the sample reported that they did not consistently wash their hands before handling their contact lenses.

Examples 13.9 and 13.10: Blood Type O

Question

Suppose the population proportion of people with blood type O is $p = 0.45$. A random sample of $n = 1000$ records is selected from a large national registry. What is the probability that the sample proportion with blood type O is 0.42 or lower?

Step 1 – Identify the Sampling Distribution: The centre is

$$\mu_{\hat{p}} = p = 0.45$$

The standard deviation is

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.45(0.55)}{1000}} \approx 0.01573$$

The expected counts are

$$np = 1000(0.45) = 450$$

$$n(1-p) = 1000(0.55) = 550$$

Both are greater than 10 so a Normal approximation is appropriate.

Examples 13.9 and 13.10: Blood Type O

Step 2 – Express the Probability: The probability of interest is

$$P(\hat{p} \leq 0.42)$$

Step 3 – Standardize: The corresponding standardized value is

$$z = \frac{\hat{p} - p}{\sigma_{\hat{p}}} = \frac{0.42 - 0.45}{0.01573} \approx -1.907$$

Step 4 – Calculate and Interpret: Using the standard Normal distribution

$$P(\hat{p} \leq 0.42) = P(Z \leq -1.907) \approx 0.0283$$

There is approximately a 2.83% probability that a random sample of 1000 records contains a sample proportion of 0.42 or lower when the population proportion is 0.45.

Examples 13.9 and 13.10: Blood Type O

R Code

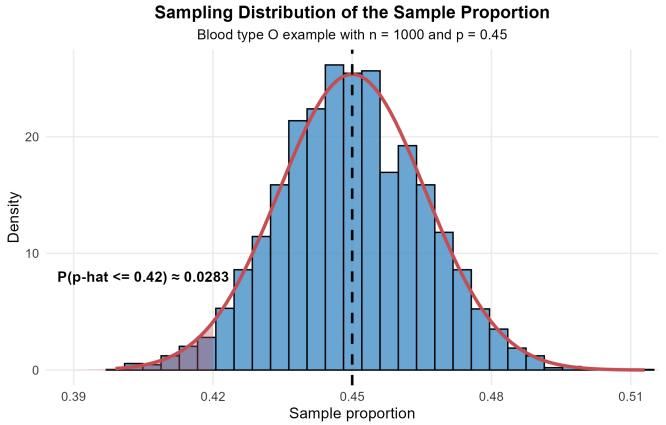
```
# Population proportion and sample size
p <- 0.45
n <- 1000

# Standard deviation of the sampling distribution
sd_p_hat <- sqrt(p * (1 - p) / n)

# Calculate P(p-hat <= 0.42)
pnorm(q = 0.42, mean = p, sd = sd_p_hat)

# [1] 0.02826514
```

Sampling Distribution of $\hat{p}$



The graph represents the possible values of $\hat{p}$ across repeated random samples of size 1000.

- The distribution is centred at $p = 0.45$
- Its standard deviation is approximately 0.01573
- The shaded region represents sample proportions at or below 0.42

Shaded Area

The area of the shaded region is $P(\hat{p} \leq 0.42) \approx 0.0283$

The Law of Large Numbers

Law of Large Numbers

For independent observations from a population with a finite mean μ, the sample mean approaches the population mean as the sample size increases

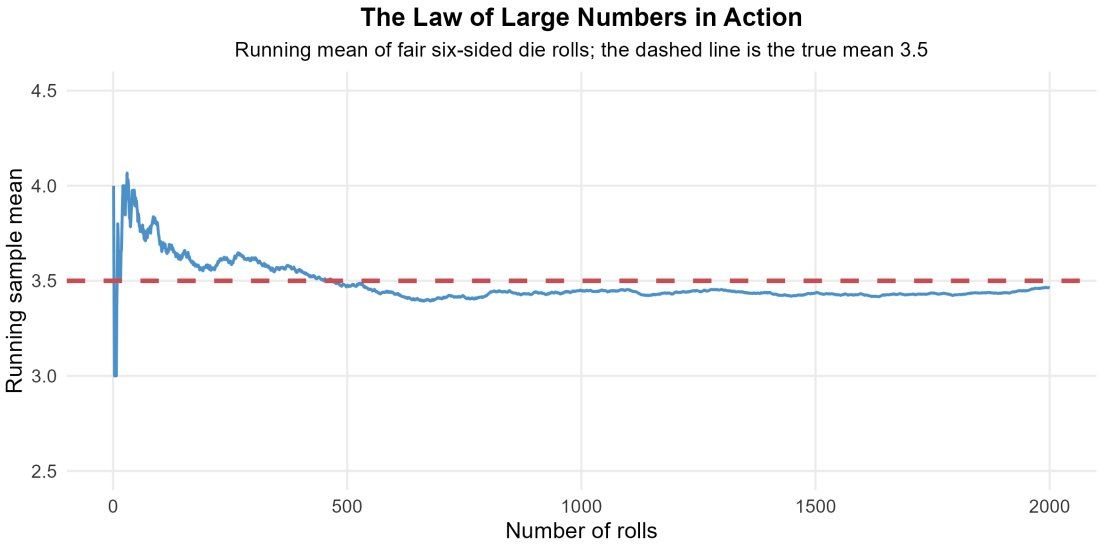
$$\bar{x}_n \longrightarrow \mu$$

For independent Bernoulli observations with a common probability p, the sample proportion approaches the population proportion

$$\hat{p}_n \longrightarrow p$$

Note: The Law of Large Numbers describes an overall long-run tendency. It does not state that each additional observation must move the statistic closer to the parameter.

The Law of Large Numbers in Action



Why the Law of Large Numbers Matters

The Law of Large Numbers provides a mathematical explanation for why averages and proportions tend to become more stable as the number of observations increases.

This long run stability is relevant in settings such as:

- Summarizing repeated physical or biological measurements and/or simulations
- Estimating average claim costs across a large insurance portfolio

Connection to Sampling Distributions

The Law of Large Numbers describes what happens as the sample size becomes very large. A sampling distribution describes how much a statistic varies for a particular finite sample size.

Example 13.11: Repeated Laboratory Measurements

A laboratory repeatedly measures the protein concentration of a reference sample. The reference concentration is 50 micrograms per millilitre. Suppose each measurement can be represented as

$$M_i = 50 + \varepsilon_i$$

where the measurement errors ε_i are independent, have mean 0 and have standard deviation 0.8 micrograms per millilitre. What happens to the running mean

$$\bar{M}_n = \frac{M_1 + M_2 + \cdots + M_n}{n}$$

as the number of measurements increases?

Example 13.11: Repeated Laboratory Measurements

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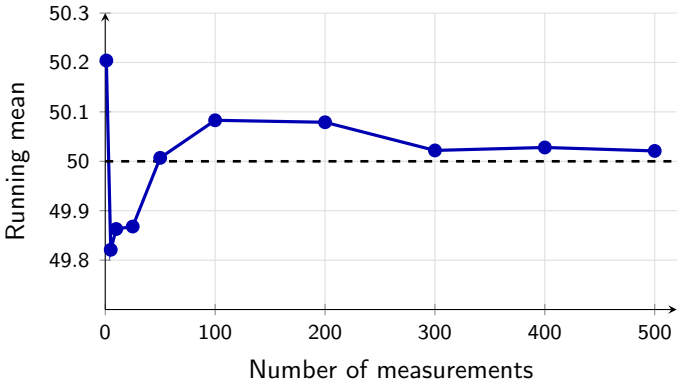
Solution

Because the measurement process is centred at 50

$$\bar{M}_n \longrightarrow 50$$

as n increases.

Example 13.11: Repeated Laboratory Measurements



Each point represents the mean of the first n simulated measurements.

- The first measurement was 50.204
- After 5 measurements, the running mean was 49.821
- After 500 measurements, it was 50.021

The running mean does not move steadily toward 50, but its fluctuations become smaller as more measurements are included.

Example 13.11: Repeated Laboratory Measurements

For independent measurements with mean 50 and standard deviation 0.8

$$E(\bar{M}_n) = 50$$

and

$$\sigma_{\bar{M}_n} = \frac{0.8}{\sqrt{n}}$$

For $n = 25$ measurements

$$\sigma_{\bar{M}_{25}} = \frac{0.8}{\sqrt{25}} = 0.16$$

For $n = 500$ measurements

$$\sigma_{\bar{M}_{500}} = \frac{0.8}{\sqrt{500}} \approx 0.0358$$

Averaging more measurements reduces the effect of random measurement error. The exact sequence will differ from one simulation to another, but the running mean tends to stabilize near 50.

Example 13.12: Claim Costs in an Insurance Portfolio

An insurance portfolio is a collection of policies. Consider the following illustrative model for the annual claim cost of one automobile policy.

Annual outcome	Probability	Claim cost
No claim	0.85	\$0
Minor claim	0.10	\$2000
Major claim	0.04	\$8000
High-cost claim	0.01	\$50000

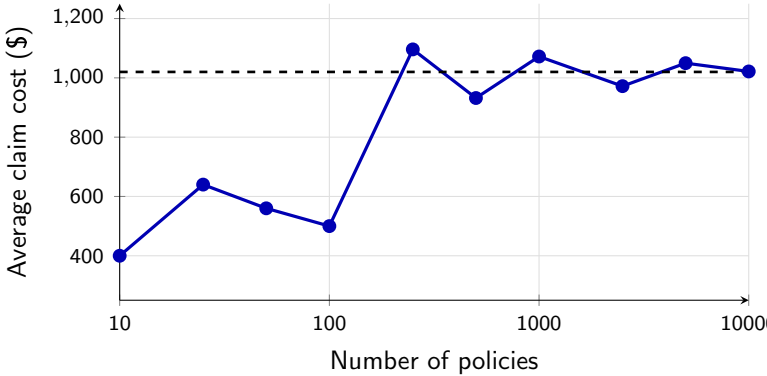
The expected annual claim cost per policy is

$$\mu = 0.85(0) + 0.10(2000) + 0.04(8000) + 0.01(50000) = 200 + 320 + 500 = \$1020$$

Question

What happens to the average annual claim cost per policy as the number of policies in the portfolio increases?

Example 13.12: Claim Costs in an Insurance Portfolio



The dashed line represents the expected claim cost of $1,020 per policy.

- Small portfolios can have unstable averages
- One high-cost claim can noticeably change the average
- The average becomes more stable as the portfolio grows

For 10000 simulated policies, the average claim cost was $1021.60

Example 13.12: Claim Costs in an Insurance Portfolio

Let C_i represent the annual claim cost for policy i. Under the illustrative model,

$$E(C_i) = \$1020$$

If the policy outcomes are independent and follow the same probability model, the Law of Large Numbers gives

$$\bar{C}_n \longrightarrow \$1020$$

as the number of policies increases.

Risk Pooling

The cost of an individual policy remains uncertain. However, the average cost per policy becomes more stable when costs are averaged across a large portfolio.

Example 13.12: Claim Costs in an Insurance Portfolio

Important Limitations

- The Law of Large Numbers does not guarantee that the average will equal \$1020 in a particular year
- Policy outcomes may not be independent when many claims result from the same storm, wildfire or other shared event
- A large portfolio does not correct an inaccurate probability model or systematic changes in claim patterns

The expected claim cost is only one component of an insurance premium. Administrative expenses, uncertainty and other financial considerations may also be included.